

**Bridging Neuroscience and Artificial Intelligence**

Laboratory Exercise A03

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# **Bridging Neuroscience and Artificial Intelligence**

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# 1. Neural Network Comparison Analysis

## Biological vs. Artificial Neurons

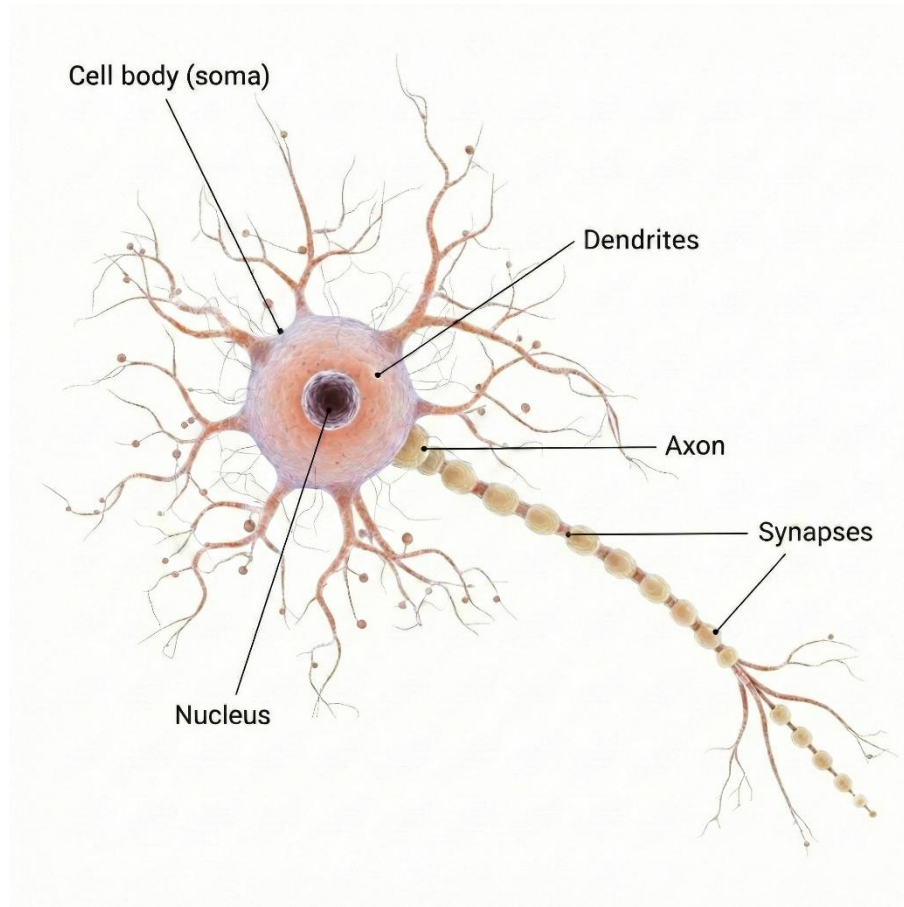


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Biological neurons consist of dendrites, a cell body (soma), an axon, and synapses. Dendrites receive incoming signals from other neurons, which are integrated in the soma. If the combined signal exceeds a threshold, an electrical impulse travels down the axon and is transmitted to other neurons through synapses (Wolfram Alpha, n.d.-a). Artificial neurons are inspired by this structure but simplify it significantly. Inputs are weighted, summed, and passed through an activation function to produce an output (Goodfellow et al., 2016).

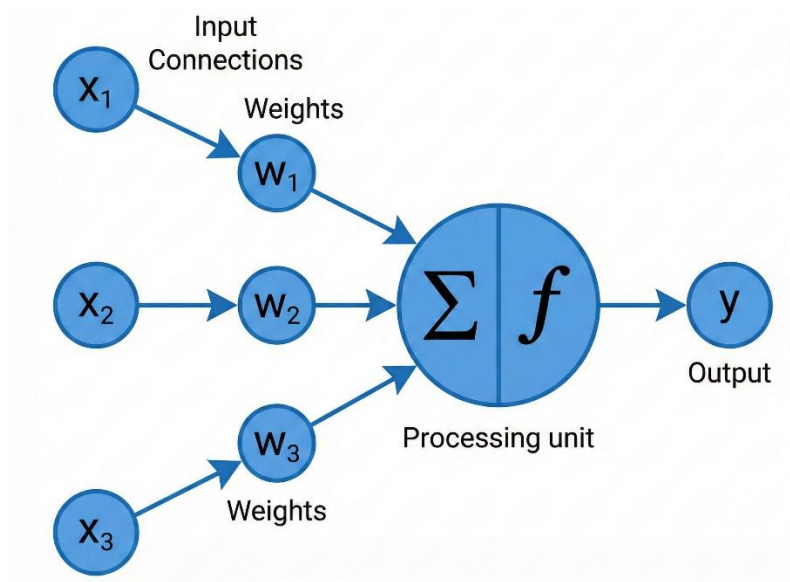


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|                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Input interpretation                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| neural cell (human)                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Alternate names                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| nerve cell   neuron   neurone                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Functional description                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| consists of a cell body, an axon, and dendrites and serves to receive, conduct, and transmit impulses in the nervous system |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Hierarchy relationships                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| anatomical groups                                                                                                           | neural cell   somatic cell                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| structural groups                                                                                                           | portion of neural tissue                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| regional parts                                                                                                              | axon   dendrite   cell body of neuron                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| members                                                                                                                     | motor neuron   set of neuroglial cells   afferent neuron   Purkinje neuron   vestibular sensory cell   dense-core granulated cell   glioblast   autonomic ganglion neuron   peripheral neuron   central neuron   cerebellar neuron   neuroblast   gabaergic neuron   nitrenergic neuron   bipolar neuron   set of dopaminergic cells   multipolar neuron   neuroendocrine cell   unipolar neuron   preganglionic neuron   pseudo-unipolar neuron   retinal pigment epithelial cell   cholinergic neuron   neuron of retina (total: 24) |
| bilateral components                                                                                                        | neural cell                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| developmental origin                                                                                                        | ectodermal cell                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Typical number in the human body                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 100 billion                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

Query from WolframAlpha

Biological inspiration strongly influenced artificial neural network design by introducing the idea of distributed processing and weighted connections (McCulloch & Pitts, 1943). However, biological neurons operate using electrochemical signals, adapt continuously, and function asynchronously, while artificial neurons rely on mathematical operations executed synchronously on digital hardware.

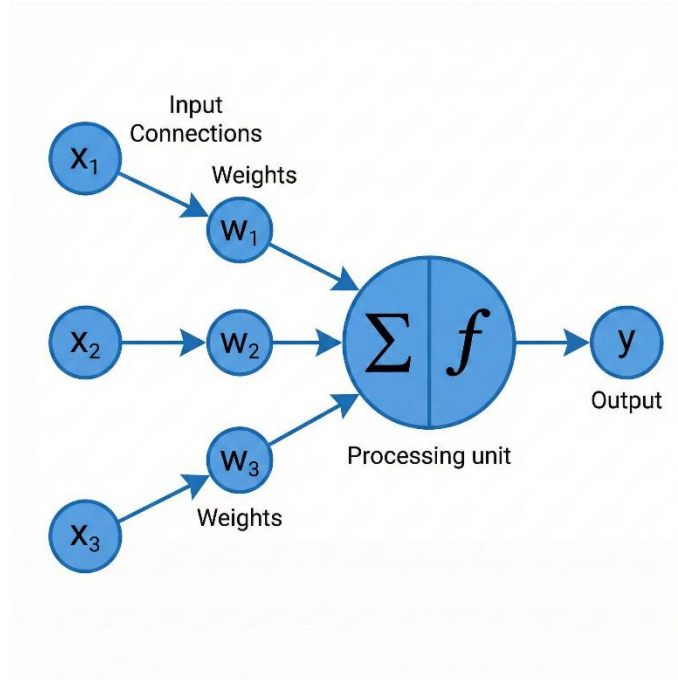


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Biological neurons excel in adaptability, energy efficiency, and robustness, but are slow and difficult to model precisely. Artificial neurons are fast, scalable, and easier to optimize but lack true autonomy, emotional context, and biological plasticity (Hawkins, 2021).

### Comparative Analysis Table

| Biological Neural Network | Artificial Neural Network | AI Architecture Example |
|---------------------------|---------------------------|-------------------------|
| Dendrites receive signals | Input nodes               | CNN input layers        |
| Soma integrates signals   | Weighted summation        | Dense layers            |
| Axon transmits output     | Output node               | RNN hidden states       |
| Synaptic strength varies  | Learnable weights         | Backpropagation         |
| Plasticity over time      | Parameter updates         | Gradient descent        |
| Parallel processing       | Vectorized computation    | GPU-based training      |

2. Brain Architecture and AI Design

Hierarchical Brain Organization

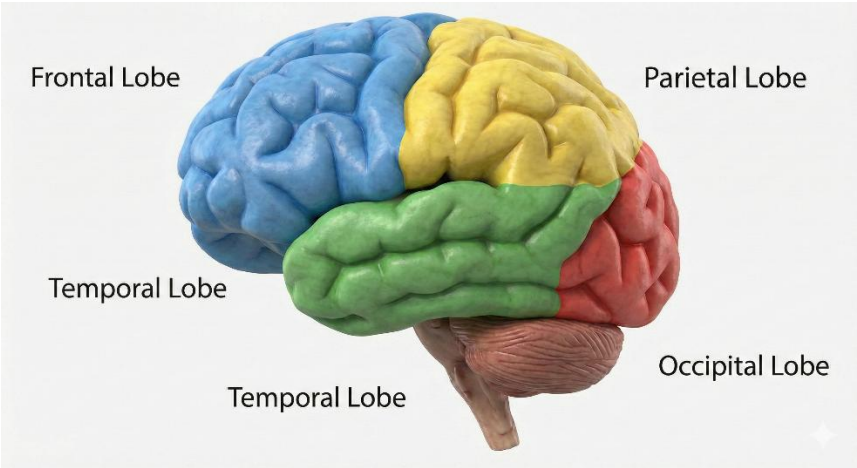


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The human brain is hierarchically organized, with specialized regions responsible for perception, memory, emotion, and decision-making (Wolfram Alpha, n.d.-b). Sensory cortices process raw input, association areas integrate information, and higher-order regions perform reasoning and planning.

|                                                                        |                                                                         |
|------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Input interpretation:                                                  |                                                                         |
| brain (human)                                                          |                                                                         |
| Latin name:                                                            |                                                                         |
| encephalon                                                             |                                                                         |
| Functional description:                                                |                                                                         |
| integrates, processes, and coordinates sensory data and motor commands |                                                                         |
| Hierarchy relationships:                                               |                                                                         |
| systemic groups                                                        | nervous system                                                          |
| anatomical groups                                                      | organ   segment of brain   nervous system of head   segment of neuraxis |
| anatomical regions                                                     | neuraxis                                                                |
| structural groups                                                      | head   head proper   neuraxis   cranial compartment                     |
| anatomical spaces                                                      | cranial cavity                                                          |
| regional parts                                                         | brainstem   midbrain   hindbrain   forebrain                            |

Query from WolframAlpha



Query from WolframAlpha

This organization closely parallels modern AI architectures. Convolutional Neural Networks (CNNs) process visual data in layered stages similar to the visual cortex, extracting edges, shapes, and objects progressively (LeCun et al., 2015). Long Short-Term Memory (LSTM) networks resemble hippocampal memory functions by retaining and updating information over time (Hochreiter & Schmidhuber, 1997). Attention mechanisms in transformer models mirror biological attention systems by prioritizing relevant information while suppressing noise (Vaswani et al., 2017).

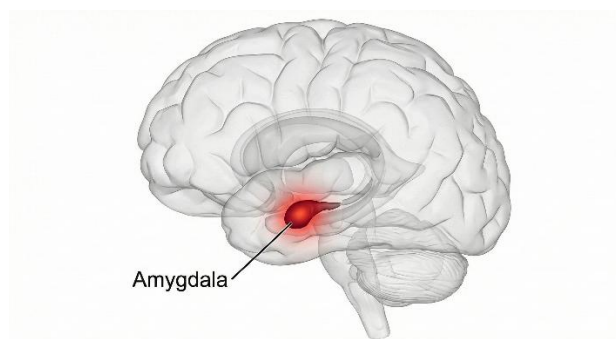


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## Amygdala and Emotion Processing

The amygdala plays a critical role in emotional processing, particularly fear, reward evaluation, and risk assessment (LeDoux, 2000). Incorporating emotion-inspired mechanisms into AI could enhance decision-making by allowing systems to weigh uncertainty, urgency, and contextual importance.

Implications include improved risk assessment in healthcare AI, better pattern recognition under uncertainty, adaptive learning based on feedback salience, and more human-aligned decision strategies (Damasio, 1994).

### 3. Integration Analysis and Future Implications

Understanding biological neural networks offers valuable insights for improving AI design.

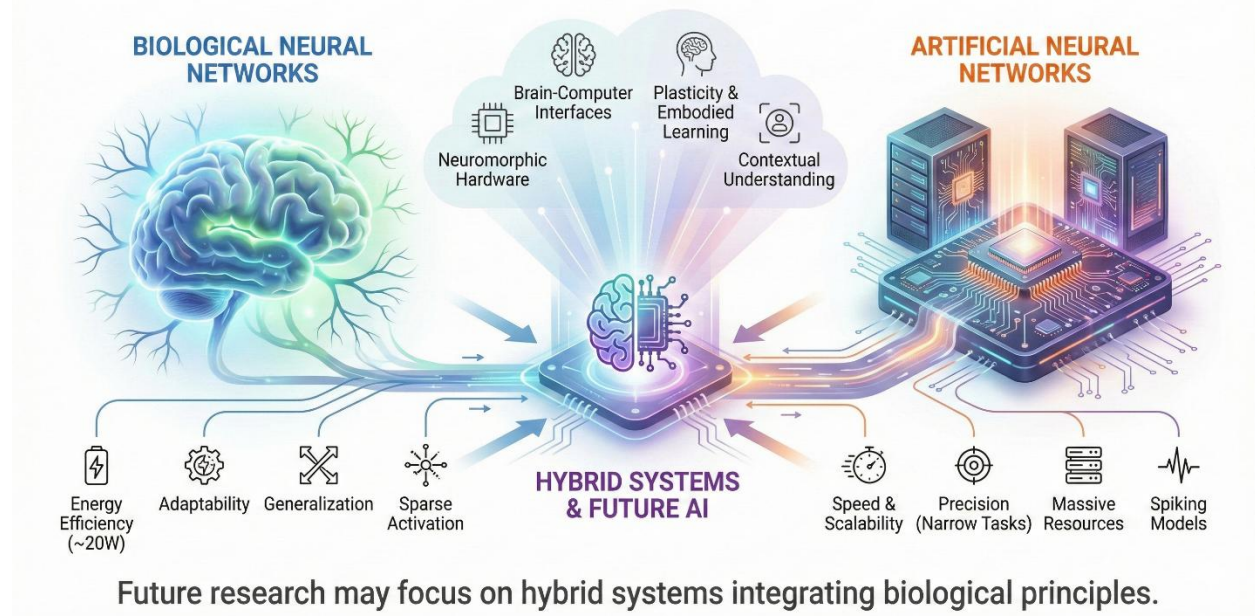


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Understanding biological neural networks offers valuable insights for improving AI design. Biological systems demonstrate efficient learning through sparse activation, local



learning rules, and continuous adaptation (Hawkins, 2021). These principles could inspire new architectures that reduce data requirements, improve generalization, and enhance robustness. Neuromorphic computing and spiking neural networks already attempt to replicate biological timing and energy efficiency (Indiveri & Liu, 2015).

Comparatively, biological systems outperform artificial ones in adaptability, energy consumption, and generalization across tasks. The human brain operates on approximately 20 watts of power, while large AI models require massive computational resources (Laughlin & Sejnowski, 2003). However, artificial systems excel in speed, scalability, and precision for narrow tasks.

Future research may focus on hybrid systems that integrate biological principles such as plasticity, emotional modulation, and embodied learning into artificial frameworks. Advances in brain-computer interfaces and neuromorphic hardware could further blur the boundary between biological and artificial intelligence. By leveraging biological insights, AI systems may overcome current limitations in flexibility, contextual understanding, and autonomous learning.

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